

**SMART INDIA
HACKATHON
2026**

A stylized illustration of a brain, split vertically. The left half is orange and filled with a circuit board pattern. The right half is green and filled with white binary code (0s and 1s). The brain is centered within a large, light gray hexagon. Below the brain, the letters 'SIH' are written in a bold, dark gray font. To the left of the brain, the word 'PAGE' is written in a large, bold, black font. Below 'PAGE', the words 'NLP', 'y &', 'DIL's', and 'and' are partially visible, suggesting a title or subtitle. The background consists of several overlapping hexagons in shades of gray.

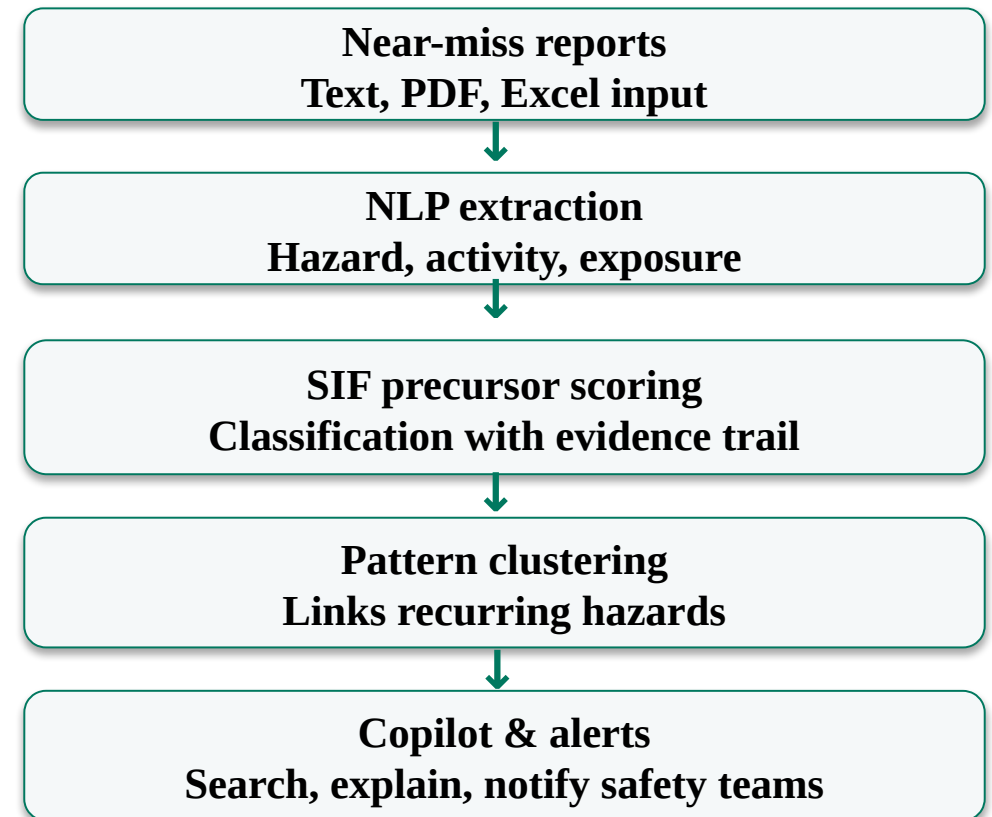
- **Problem Statement ID – SIH26165**
- **Problem Statement Title-AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports**
- **Theme-Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Sable**

SIF SENTINEL-Turn unstructured near-miss reports into explainable SIF intelligence.

Idea:

- **SIF Sentinel** turns unstructured near-miss reports into evidence-backed SIF (Serious Injury/Fatality) precursor intelligence.
- Extracts hazard, activity, and exposure from raw report text — no manual tagging.
- Scores each report's SIF potential and shows the exact evidence behind the score, not a black-box number.
- Clusters similar near-misses across reports and sites to surface recurring hazard patterns before they escalate.
- A copilot lets safety teams query, compare, and act on flagged reports in plain language.

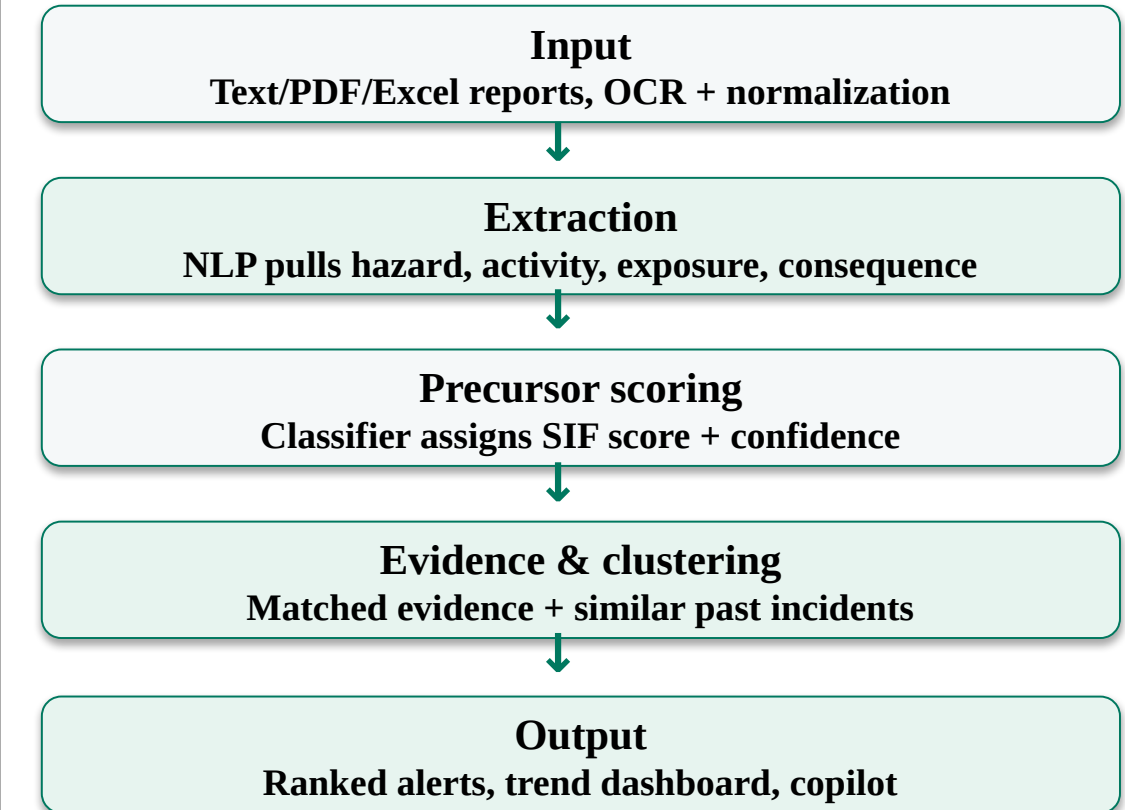
Workflow Diagram:



TECHNOLOGIES

- **NLP/NER** — spaCy pipeline fine-tuned on a custom oil & gas safety lexicon (permit-to-work, LOTO, confined space, hot work, H2S) — not a generic off-the-shelf model
- **SIF classification** — weak-supervision labeling (rule-based + LLM-assisted, validated against public CSB/OSHA incident investigation reports) feeding a lightweight classifier — addresses the "no labeled data" problem head-on instead of hiding it
- **Explainability** — evidence retrieval: each flagged report is paired with the specific extracted hazard/exposure terms and similar past incidents that drove the score
- **Similarity & clustering** — sentence embeddings + density-based clustering (HDBSCAN) to group recurring hazard patterns
- **Copilot** — RAG grounded strictly in retrieved reports and evidence (no ungrounded generation)
- **App/infra** — React, FastAPI, PostgreSQL + pgvector, Docker

TECHNICAL APPROACH



FEASIBLE

- Historical near-miss / incident data
- Public safety datasets + annotated samples
- Mature NLP / ML frameworks
- Modular, API-driven deployment

KEY CHALLENGES

- Limited labelled SIF precursor data
- Oil & gas terminology and abbreviations
- False positives can cause alert fatigue
- Sensitive industrial incident information

MITIGATION

- Taxonomy + expert annotation + weak supervision
- Domain vocabulary + model fine-tuning
- Confidence thresholds + human review
- Anonymisation + access control + encryption

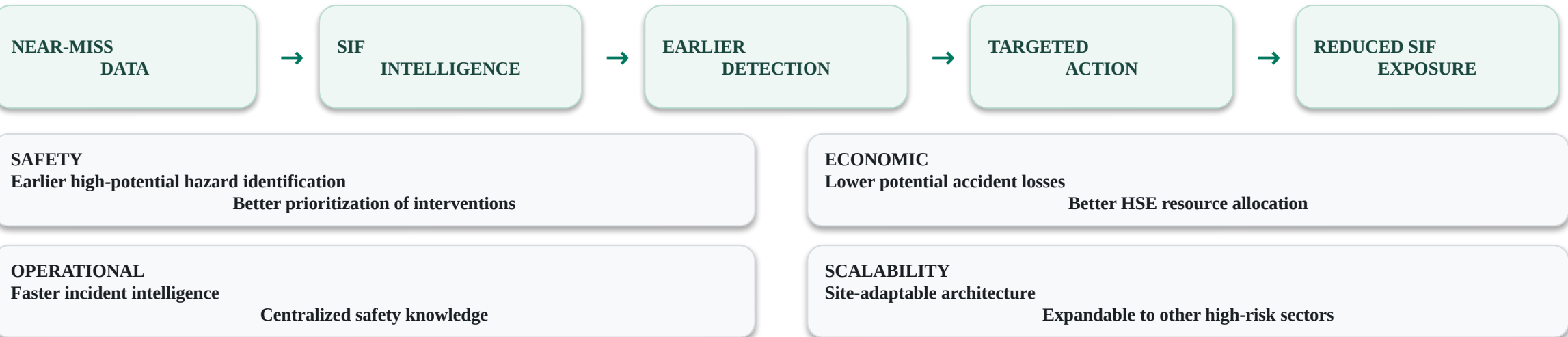
Who benefits?

SAFETY PROFESSIONALS
Prioritize high-potential reports
instead of reviewing everything equally.

OPERATIONS / PLANT
Identify recurring hazards
and emerging risk trends.

HSE TEAMS
Move from reactive analysis
to proactive prevention.

IMPACT CHAIN



Every near miss becomes an opportunity to prevent the next serious incident.

SafeOCS — Precursor Safety Data Program

US Bureau of Transportation Statistics / BSEE. Voluntary near-miss and unsafe-act/condition reporting for offshore oil & gas. [safeocs.gov](https://www.safeocs.gov)

IOGP Safety Performance Indicators

International Association of Oil & Gas Producers. Global incident database with causal factors since 1985. [iogp.org](https://www.iogp.org)

Automatically Learning Construction Injury Precursors from Text

90,000+ oil & gas incident reports (title, description, root cause) with severity/injury labels. [arXiv:1907.11769](https://arxiv.org/abs/1907.11769)

Expert-Annotated High-Pressure Gas Incident Dataset

NER + cause-extraction annotations on petrochemical/oil-refining incident reports. [arXiv:2310.12074](https://arxiv.org/abs/2310.12074)

Petrobras 3W Dataset

Public benchmark of rare undesirable events in oil wells for anomaly detection/diagnosis. github.com/petrobras/3W

Open HSE Reference Datasets (SmartQHSE)

IOGP Life-Saving Rules, HAZOP guidewords, incident taxonomy — for hazard vocabulary design. smartqhse.com/datasets